

Traffic Light Optimization using SUMO at the Samsat Intersection*

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Abstract—Traffic congestion is a common problem in metropolitan areas, with vehicles on the road increasing annually. The queue of vehicles at intersections contributes to traffic congestion, highlighting the need for adaptive traffic light systems to prevent long queues. In this study, we present a simulation of adaptive traffic light system at the Samsat intersection in Bandung, Indonesia. The simulation model of Samsat intersections developed using SUMO and based on actual data. The parameters used as input for the simulation are the duration of the green light plan, the number of vehicles based on their class and the simulation time. To verify the model, the data from the SUMO simulation is compared with the actual data from the Area Traffic Control System (ATCS) in Bandung. Therefore, the comparison results show that the SUMO simulation reflects the true behavior of the traffic given the assumption of no manual overrides.

Index Terms—adaptive traffic light, traffic jam, SUMO

I. INTRODUCTION

The growth in the number of vehicles occurs significantly every year. In Indonesia, Based on data from the Central Statistics Agency, the number of vehicles in Indonesia was 126,508 million in 2018. This number increased in 2019, with 133.617 million vehicles. The trend continued in 2020, with a recorded total of 136.137 million vehicles [1]. In addition, people tend to prefer using private vehicles rather than using public transportation, which indirectly increases the number of vehicles each year. Based on Figure 1, it can be seen that the number of vehicles continues to increase every year in each class. This number is predicted to continue to grow every year [2]. The habit of Indonesian citizens of using private vehicles more often than public transportation is considered as one of

the main causes of the increasing number of vehicles on the road. The increase of vehicles on the road will lead to a new problem, such as traffic jams. [3].

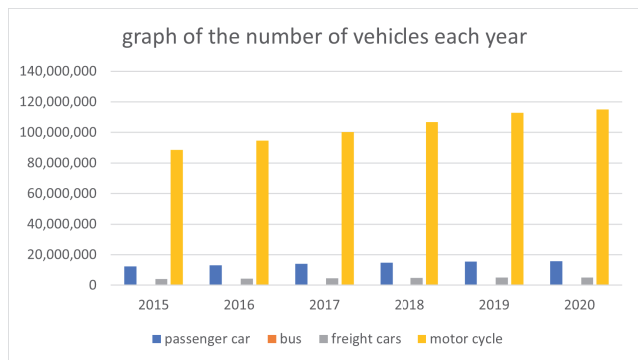


Fig. 1. Number of vehicles each year.

Traffic jams mainly occur due to the long queue of vehicles and the number of intersections. The application of traffic lights is considered a solution to break down traffic jams. However, the green light delay time in the installed traffic lights is usually not adjusted to the length of queue. Moreover, with the increasing number of vehicles entering an intersection, especially during busy times, the use of traffic lights become inefficient in operating the traffic network, causing congestion. [4]. To optimize delays, several researchers proposed an adaptive traffic light. The adaptive traffic light calculates the delay time based on the number of vehicles entering each road

segment. The adaptive traffic light requires several supporting parameters: a vehicle number detector, average data speed based on vehicle type, and road segment data. This system has been applied in developed countries, with several traffic system in a city. [5].

Integration between traffic lights is essential to determine traffic engineering policies under certain conditions. To create an adaptive traffic light, simulation is needed to get the traffic flow at an intersection. In this research, we simulate the traffic light at Samsat intersection, Buah Batu district, West Java, Indonesia. The intersection was chosen because the number of CCTV is the most [9]. In addition, Samsat intersection has the longest waiting time in Indonesia [6] [2]. Samsat intersection as shown in Figure 2 where looks very congested. The SUMO simulation model is created using a real data, obtained by the ATCS owned by Bandung Transportation Service. To simulate the traffic lights, we conducted a field study at the intersection to observe the currently implemented system. [7].

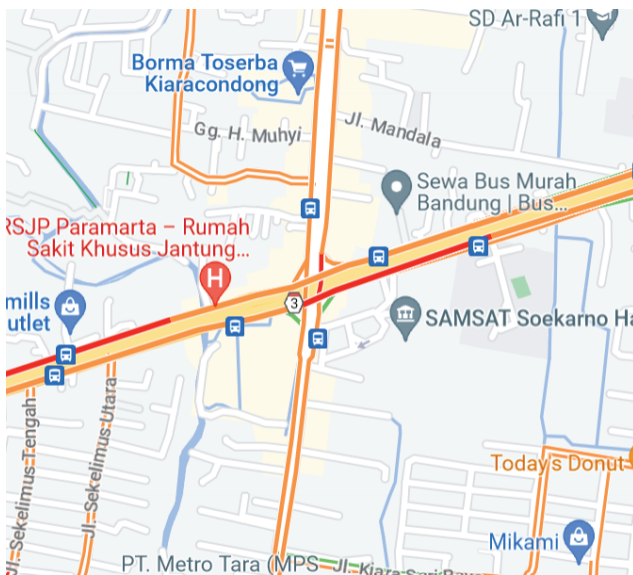


Fig. 2. Samsat intersection

II. RELATED WORKS

With the increasing population and the subsequent rise in vehicles on the road, urban areas are experiencing significant traffic congestion. Intelligent Transportation Systems (ITS) have been proposed as a solution, leveraging existing infrastructure to address this issue. However, the strategies to tackle urban traffic problems require thorough testing before implementation.

Conducting real-world testing for traffic management techniques can be expensive and impractical, leading to the widespread use of traffic simulators to test and enhance these techniques virtually. These simulators make use of real-world maps and real-time traffic information to replicate the complex dynamics of urban traffic.

To achieve accurate simulation results, it becomes crucial to import high-quality, real-world maps into the simulator. Unfortunately, maps obtained from sources like open street maps/osmWebWizard may contain erroneous or faulty information. Addressing these inaccuracies before running simulations is essential to ensure realistic and reliable results.

Researchers in [3] proposed a detailed procedure for importing real-world maps from osmWebWizard and modifying their internal elements to replicate the real world for simulation purposes accurately. Their work, titled "SUMO-Network: Generating and Remodeling Real World Map using osmWebWizard and NETEDIT," presented a comprehensive approach to address the challenges of map importation and modification. By applying their procedure, the authors successfully eliminated unrealistic simulation results caused by faulty map data. This research provides valuable insights into the crucial step of map preparation in traffic simulation studies [3].

Several research studies have focused on the simulation and optimization of traffic control systems using the SUMO (Simulation of Urban Mobility) platform. These studies explore various aspects of traffic management, route optimization, and simulation scenarios. The following papers provide valuable insights into the related work in this field.

Researchers in [8] propose a framework for intelligent traffic control using multiagent learning within the SUMO-based simulation environment. Their work aims to enhance traffic management in urban areas by leveraging intelligent transportation systems [8].

Researchers in [9] discuss the implementation of SUMO and OpenStreetMap (OSM) in simulating urban mobility. They highlight the importance of incorporating accurate and up-to-date map data to achieve realistic simulation results [9].

Researchers in [6] explores reinforcement learning techniques for vehicle route optimization within the SUMO environment. Their research aims to improve traffic efficiency and reduce congestion through intelligent route selection [6].

Researchers in [5] focus on modelling road traffic flow using the SUMO simulator. Their study contributes to understanding the dynamics of traffic behaviour and provides insights for effective traffic management strategies [5].

Researchers in [7] evaluate the impact of traffic congestion using SUMO. Their research investigates the consequences of congestion on traffic flow and explores potential solutions for alleviating congestion-related issues [7].

Researchers in [4] propose embedding actual vehicles in SUMO for large-scale Intelligent Transportation System (ITS) scenario emulation. Their work enables realistic simulation of ITS scenarios and facilitates the evaluation of traffic control strategies [4].

Researchers in [2] present an adaptive method for traffic signal control based on fuzzy logic using the SUMO traffic simulator. Their research focuses on optimizing traffic signal timing to improve traffic flow efficiency [2].

Researchers in [10] develop a high-accuracy and large-scale traffic simulation scenario using a novel traffic demand

calibration method in SUMO. Their work provides a realistic simulation environment for evaluating traffic management strategies [10].

Researchers in [4] propose a large-scale SUMO-based emulation platform for intelligent transportation systems. Their research offers a comprehensive framework for simulating and evaluating traffic control strategies on a large scale [4].

Researchers in [11] conduct a systematic literature review on reinforcement learning in urban network traffic signal control. Their study provides an overview of existing research, identifies key challenges, and highlights potential directions for future work [11].

Several studies have used SUMO for traffic simulations and applied reinforcement learning to SUMO simulations. In this study we create the sumo simulation based on actual data.

III. METHODS

To generate a SUMO simulation model, we obtained the real data from ATCS Bandung. The simulated model is then compared with the real data using the research flowchart as shown in Figure 3.

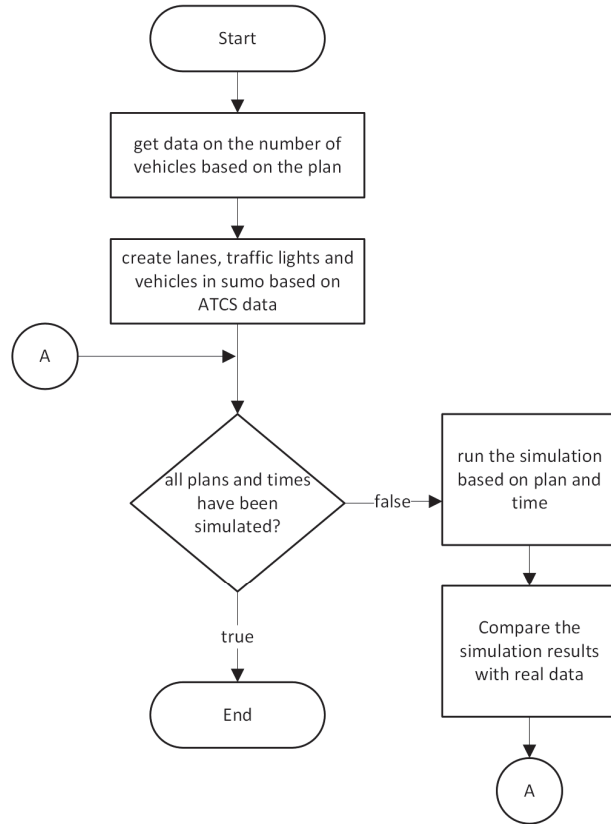


Fig. 3. Research flowchart

A. Plan in Samsat Intersection

Traffic light at Samsat intersection is divided based on a specific time, with each time having a different time setting

for each road segment. This time setting is referred to as a "plan". The plans as shown in Table 1.

[h]

TABLE I
PLAN AT SAMSAT INTERSECTION

Plan Number	Green Time (s)			
	East	West	South	North
1	180	70	50	90
2	170	75	55	75
3	150	85	55	85
4	80	65	40	60
5	60	40	25	40
6	40	25	15	25
7	55	30	20	30
8	38	20	20	20

Table 1 shows that each plan has a different green light duration, while the red light duration for one road segment is a calculation of the green and yellow lights in another road segment. The cycle runs with the duration of a green light by the plan, then continues with a yellow light for 3 seconds and a red light for 2 seconds before moving on to a green light for another road segment.

B. SUMO Configuration

SUMO (Simulation of Urban Mobility) is a traffic simulation software used for modeling and simulating urban traffic and transportation systems. It allows researchers and planners to analyze traffic flow, congestion, and various transportation scenarios to make informed decisions for improving traffic management and urban mobility. Setting up SUMO begins with creating a net using the web wizard by capturing Samsat intersection. After that, we modified the web wizard obtained by changing the road segment and cycle data based on actual data. The results of the process as shown in Figure 4.

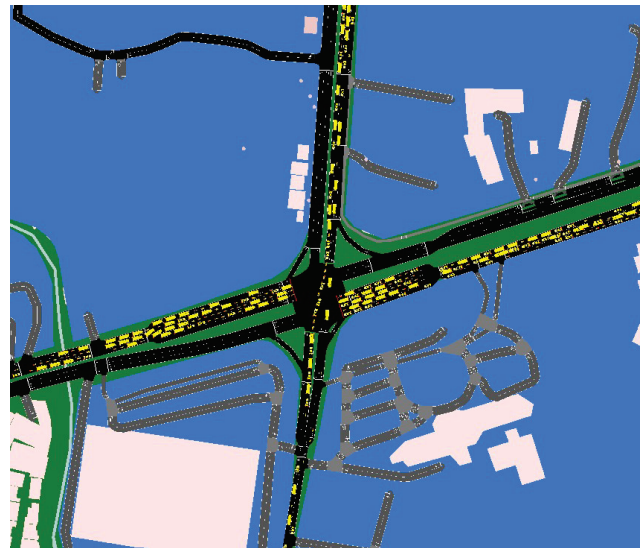


Fig. 4. The result of creating a net in SUMO.

C. Vehicle data

The data used in this study uses data obtained from CCTV cameras that have been installed at each foot of the intersection. Object detection is then carried out from CCTV cameras using the YOLO v4 method [12]. The objects detected are motorcycles, cars, and large vehicles such as buses and trucks. This detection system is implemented at each intersection segment using Jetson Nano. Since we used edge computing for each intersection segment, YOLO v4 is reliable for computation limitation. The output of this system is the number of each vehicle, the average speed of each vehicle, and the gap between vehicles at the end of the intersection segment every 5 minutes on working days. The vehicle data used in this simulation is data for one week from April 1 to April 7, 2023.

D. Simulation

Before doing the simulation, we created a program to generate the trip for each vehicle that will enter the simulation. The program is called a "trip generator". In this program, it will generate vehicle data and distribute it based on the time and according to the plan. Based on these data, the simulation will generate the mean of travel time, mean of waiting time, average speed, and departure delay. The simulation continues for all plans for one week.

IV. RESULT AND DISCUSSION

A. Simulation result

The simulation results are divided per day based on the change in plan on that day. The number of vehicles that were initially per hour changed to per plan used on that day. Samsat intersection on monday has eight changes of plan, with the results shown in Table 2.

[h]
TABLE II
MONDAY SIMULATION RESULTS

Monday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	41.23	753.8	305.97	967.65
05:01-06:00	5	38.55	864.6	305.97	946
06:01-08:00	4	10.74	1565.91	3568.54	5489.52
08:01-13:00	2	20.6	1278.4	2762.2	4651.7
13:01-18:00	3	15.2	1623.8	2975.1	6103.2
18:01-21:00	4	17.3	1524.7	3181.5	5539.6
21:01-23:00	5	28.55	755.6	310.54	875.7
23:01-23:59	6	29.06	754.02	318.66	831.09

Table 2 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 2 are from 06:01 to 08:00 with the longest waiting time, travel time, and depart delay. Samsat intersection on tuesday has nine changes of plan, with the results shown in Table 3.

Table 3 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 3 are from 18:30 to 21:00 with the longest

[h]
TABLE III
TUESDAY SIMULATION RESULTS

Tuesday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	43.23	768.8	302.97	965.65
05:01-06:00	7	41.04	852.6	311.93	855
06:01-08:00	1	12.53	903.2	3287	4321
08:01-12:00	2	25.6	902	2852	3210
12:01-16:00	3	24.1	905	2672	3202
16:01-18:30	2	11.1	908	4783	5490
18:31-21:00	4	21.1	908	4777	5225
21:01-22:00	5	26	906	1060	2053
22:01-23:59	6	28	904	955	902

waiting time, travel time, and depart delay. Samsat intersection on wednesday has eight changes of plan, with the results shown in Table 4.

[h]
TABLE IV
WEDNESDAY SIMULATION RESULTS

Wednesday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	24	865	788	889
05:01-05:30	5	22	862	790	910
05:31-12:00	1	13	979	3820	5098
12:01-12:30	2	22	901	2020	3018
12:31-16:00	3	13	911	1209	2076
16:01-21:00	4	8.5	901	4065	6181
21:01-22:00	5	21	855	854	1276
23:01-23:59	6	23	801	832	907

Table 4 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 4 are from 16:01 to 21:00 with the longest waiting time, travel time, and depart delay. Samsat intersection on thursday has nine changes of plan, with the results shown in Table 5.

[h]
TABLE V
THURSDAY SIMULATION RESULTS

Thursday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	21	821	964	1211
05:01-06:00	7	20	842	906	1365
06:01-09:00	1	9.1	1011	3890	5031
09:01-12:00	2	15	972	2541	3075
12:31-16:00	3	14	901	2021	2563
16:01-19:00	2	10.1	903	3401	5120
19:01-21:00	4	13.2	954	2056	3078
21:01-22:00	5	17	901	972	1043
22:01-23:59	6	19	855	993	1076

Table 5 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 5 are from 06:00 to 09:00 with the longest waiting time, travel time, and depart delay. Samsat intersection on friday has ten changes of plan, with the results shown in Table 6.

[h]
TABLE VI
FRIDAY SIMULATION RESULTS

Friday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	22	531	955	1031
05:01-06:00	7	21	532	904	1046
06:01-08:00	1	9.03	1011	4012	6891
08:01-12:00	3	12	911	2911	3239
12:01-13:00	2	15	901	1022	1514
13:01-14:00	3	13	903	1031	1501
14:01-19:45	4	8	1071	3891	7010
19:45-20:45	5	11	1012	3910	6950
20:45-23:00	4	18	964	2879	3261
23:00-23:59	6	19	932	1201	1510

Table 6 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 6 are from 14:01 to 19:45 with the longest waiting time, travel time, and depart delay. Samsat intersection on Saturday has eight changes of plan, with the results shown in Table 7.

[h]
TABLE VII
SATURDAY SIMULATION RESULTS

Saturday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	25	901	672	865
05:01-08:00	4	19	902	2010	2543
08:01-12:00	2	17	901	1523	1818
12:01-13:30	4	18	902	1587	2087
13:31-18:00	3	18.8	901	1798	2197
18:01-21:00	4	17	903	1886	2265
21:01-23:00	5	24	903	976	1102
23:01-23:59	6	21	865	987	1209

Table 7 shows that the waiting time, travel time, and vehicle depart delay are long, especially during peak hours. Peak hours in the Table 7 are from 05:01 to 08:00 with the longest waiting time, travel time, and depart delay. Samsat intersection on Sunday has seven changes of plan, with the results shown in Table 8.

[h]
TABLE VIII
SUNDAY SIMULATION RESULTS

Sunday					
Time	Plan	Avg speed (km/hours)	Depart delay(s)	Waiting time(s)	Travel time(s)
00:00-05:00	6	28	866	978	1011
05:01-08:00	4	21	901	1520	1820
08:01-13:00	2	19	900	1490	1716
13:01-18:00	3	15	901	1590	1876
18:01-21:00	4	14	903	1554	1765
21:01-23:00	5	20	900	902	1016
23:01-23:59	5	21	862	776	885

Table 8 shows that the waiting time, travel time, and vehicle departure delay are long, especially during peak hours. Peak hours in the Table 8 are from 05:01 to 08:00 with the longest waiting time, travel time, and depart delay.

B. Analysis

From the simulation results, it can be seen that the average waiting time, travel time and departure delay are long. some even during peak hours, the waiting time reaches as long as 2 hours. Peak hours range from 6.00-8.00 and 16.00-20.00. these hours resulted in the longest waiting time of the day. The number and type of vehicles that enter the simulation also affect the waiting time. This can be seen if a road section has a lot of large vehicles, there is a long queue. These results are in accordance with our observations at ATCS Bandung city.

V. CONCLUSION

Traffic congestion poses a significant challenge in densely populated urban areas, particularly in large cities with limited road capacity and a growing number of vehicles. In an effort to address this pressing issue, our research focused on observing and analyzing traffic conditions at the Samsat intersection in Kota Bandung, Indonesia, known for its heavy traffic flow during peak hours. To achieve this, we employed the SUMO simulator, utilizing authentic data provided by the Bandung Transportation Service to conduct a week-long simulation. This simulation allowed us to capture crucial traffic parameters such as travel times and waiting times at the intersection.

The findings from our simulation revealed that the Samsat intersection experiences substantial traffic congestion during peak hours, specifically from 5.31 AM to 12.30 PM and 4.00 PM to 9.00 PM. Notably, the highest travel time recorded in the SUMO standard simulation reached 7010 seconds, occurring on Friday between 2.01 PM and 7.45 PM. Similarly, the longest waiting time observed was 4783 seconds, documented on Tuesday between 4.10 PM and 6.30 PM. These results align with our direct observations and highlight the need for an improved traffic management plan for the intersection, as the current plan provided by the Bandung City Transportation Agency may not be fully optimized for handling varying traffic conditions.

Moving forward, we emphasize the urgency of future research endeavors focused on optimizing traffic plans that can dynamically adapt to fluctuating traffic conditions. The application of cutting-edge technologies such as machine learning, deep learning, and reinforcement learning presents promising opportunities for implementing adaptive traffic management strategies. By harnessing the power of these advanced technologies, it is possible to devise more efficient and responsive traffic management systems, leading to a smoother flow of vehicles and reduced congestion. Such adaptive traffic management approaches hold the potential to significantly alleviate the challenges posed by urban traffic congestion and enhance overall transportation efficiency and sustainability.

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